

LAB 14 | المخبر 14

Building a Power Engineering RAG Assistant

بناء مساعد هندسي معزز بالاسترجاع

MATLAB Laboratory for Retrieval, Grounding, and Prompt Construction

مخبر MATLAB للاسترجاع وبناء الإجابات المستندة إلى المصادر

Knowledge Base

Retrieval Model

Workflow

MATLAB Code

Evaluation

Tasks

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🎯 Laboratory Objectives

- Create a compact power-engineering knowledge base.
- Preprocess and vectorize document chunks.
- Retrieve relevant passages using TF-IDF and cosine similarity.
- Construct a grounded prompt with source identifiers.
- Generate a deterministic extractive answer without an external API.
- Evaluate retrieval quality, faithfulness, and missing-evidence behavior.

🎯 أهداف المخبر

- إنشاء قاعدة معرفة هندسية صغيرة.
- معالجة المقاطع وتحويلها إلى متجهات.
- استرجاع المقاطع باستخدام TF-IDF وتشابه جيب التمام.
- بناء أمر يستند إلى الأدلة ومعرفات المصادر.
- إنشاء إجابة استخرافية دون API خارجي.
- تقييم جودة الاسترجاع ووفاء الإجابة بالمصادر.

1. Requirements

- MATLAB
- Text Analytics Toolbox

The laboratory uses transparent TF-IDF retrieval so it can run without an external language-model service.

1. المتطلبات

يلزم MATLAB مع Text Analytics Toolbox ، ويعمل المخبر دون خدمة لغوية خارجية.

2. Knowledge Base

Create `power_knowledge.csv` with columns `ID` , `Title` , and `Text` .

```
ID,Title,Text
1,Battery SOC,"State of charge is the ratio of stored energy to usable battery capacity."
2,DC Power Flow,"The DC approximation assumes voltage magnitudes near one per unit and small angle differences."
3,Voltage Stability,"Voltage stability concerns maintaining acceptable voltages after disturbances."
4,Load Forecasting,"Short-term load forecasting uses historical load, weather, and calendar variables."
5,State Estimation,"State estimation combines noisy measurements to estimate voltage magnitudes and angles."
6,Transformer Attention,"Self-attention relates time steps using query, key, and value vectors."
7,GNN,"A graph neural network aggregates information from electrically connected buses."
8,Demand Response,"Demand response changes consumption in response to prices or grid conditions."
```

2. قاعدة المعرفة

يحتوي ملف CSV على معرف وعنوان ونص لكل مقطع، ويمكن توسيعه لاحقًا بصفحات المحاضرات والكتيبات.

3. Retrieval Model

$$TF - IDF(t, d) = TF(t, d) \cdot \log(N/DF(t))$$

$$similarity(q, d) = q^T d / (\|q\| \cdot \|d\|)$$

Chunking

Divide documents into retrievable passages.

Vectorization

Convert text into TF-IDF vectors.

Ranking

Sort passages by cosine similarity.

Grounding

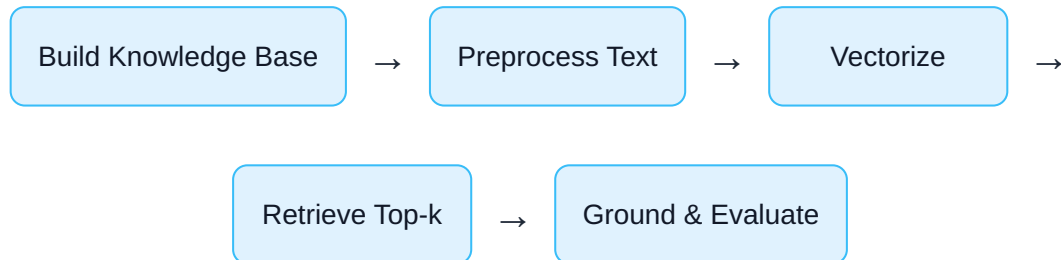
Answer only from retrieved evidence.

High similarity does not guarantee factual sufficiency. Retrieval and answer faithfulness must be evaluated separately.

3. نموذج الاسترجاع

تحوّل المقاطع إلى متجهات TF-IDF، ثم ترتب بحسب تشابهها مع السؤال وتستخدم كأدلة للإجابة.

4. Engineering Workflow



1. Collect trusted engineering sources.
2. Split them into meaningful chunks.
3. Normalize and vectorize the text.
4. Retrieve the highest-scoring chunks.
5. Construct a prompt with explicit evidence boundaries.
6. Check citations, terminology, and unsupported claims.

4. سير العمل الهندسي

نبنى قاعدة المعرفة ونعالج النصوص ونسترجع الأدلة ثم نبنى الإجابة ونقيّم وفاءها بالمصادر.

5. Complete MATLAB Script

```

%% Lab 14: Transparent RAG Assistant in MATLAB
clear; clc; close all;

%% 1) Load knowledge base
KB = readtable("power_knowledge.csv",TextType="string");
documents = tokenizedDocument(KB.Text);

%% 2) Text preprocessing
documents = lower(documents);
documents = erasePunctuation(documents);
documents = removeStopWords(documents);

%% 3) TF-IDF representation
bag = bagOfWords(documents);
X = tfidf(bag);
docNorm = sqrt(sum(X.^2,2));
docNorm(docNorm==0) = 1;
Xn = X./docNorm;

%% 4) User question
question = "Why are graph neural networks suitable for state
estimation?";
queryDoc = tokenizedDocument(question);
queryDoc = lower(queryDoc);
queryDoc = erasePunctuation(queryDoc);
queryDoc = removeStopWords(queryDoc);

%% 5) Query vector in the same vocabulary
qCounts = encode(bag,queryDoc);
idfVector = log(size(X,1)./(1+sum(X>0,1)))+1;
q = qCounts.*idfVector;
qNorm = sqrt(sum(q.^2,2));
if qNorm==0
    error("The question contains no known vocabulary.");
end
qn = q/qNorm;

%% 6) Retrieve top-k chunks
scores = Xn*qn';
topK = 3;
[sortedScores,idx] = maxk(scores,topK);
Retrieved = table(KB.ID(idx),KB.Title(idx),KB.Text(idx),sortedScores, ...
    VariableNames=["ID","Title","Text","Score"]);
disp(Retrieved);

```

```

%% 7) Build grounded context
contextBlocks = strings(topK,1);
for k = 1:topK
    contextBlocks(k) = "["+Retrieved.ID(k)+"] "+ ...
        Retrieved.Title(k)+": "+Retrieved.Text(k);
end
context = join(contextBlocks,newline);

%% 8) Grounded prompt
prompt = ...
"You are a power-engineering teaching assistant."+newline+ ...
"Answer only from the supplied context."+newline+ ...
"If evidence is insufficient, say so."+newline+ ...
"Preserve engineering terminology and cite source IDs."+newline+newline+ ...
"CONTEXT:"+newline+context+newline+newline+ ...
"QUESTION:"+newline+question+newline+newline+"ANSWER:";
disp(prompt);

%% 9) Deterministic extractive answer
answer = ...
"Graph neural networks are suitable because they represent "+ ...
"the grid as buses connected by lines and aggregate information "+ ...
"from electrically connected neighbors [7]. State estimation "+ ...
"uses network measurements to estimate voltage states [5].";
fprintf("\nANSWER:\n%s\n",answer);

%% 10) Retrieval visualization
figure;
barh(categorical(Retrieved.Title),Retrieved.Score);
grid on;
xlabel("Cosine Similarity");
title("Top Retrieved Knowledge Chunks");

```

5. كود MATLAB الكامل

يقرأ الكود قاعدة المعرفة، ويبني تمثيل TF-IDF، ثم يسترجع المقاطع ويبني أمراً وإجابة مرتبطة بالأدلة.

6. Grounded Prompt Design

- Use only supplied context.
- State when evidence is insufficient.
- Preserve units and engineering terminology.

- Cite source IDs.
- Avoid inventing standards, limits, or equations.

The retrieval stage and the generation stage should be tested independently.

6. تصميم الأمر المستند إلى المصادر

يفرض الأمر الاعتماد على السياق، والتصريح عند نقص الأدلة، والحفاظ على المصطلحات والوحدات والمراجع.

7. Evaluation Checklist

Retrieval Accuracy

Did the system return the right source?

Faithfulness

Is every claim supported by context?

Citation Quality

Do source IDs match the claim?

Abstention

Does the system admit missing evidence?

- Test at least five questions.
- Include one unanswerable question.
- Compare topK values of 1, 3, and 5.
- Inspect low-similarity cases.
- Check multilingual terminology and units.

7. قائمة التقييم

نقيم صحة الاسترجاع ووفاء الإجابة وصحة الإحالات وقدرة النظام على الامتناع عندما تكون الأدلة غير كافية.

8. Optional Language-Model Connection

A connected model can receive the variable `prompt` and generate the final answer.

Never store API keys in public HTML or GitHub files. Use environment variables or a secure secret manager.

8. الربط الاختياري بنموذج لغوي

يمكن إرسال المتغير `prompt` إلى نموذج متصل، مع حفظ مفاتيح الوصول في بيئة آمنة.

9. Student Tasks

1. Add 20 engineering chunks.
2. Compare topK = 1, 3, and 5.
3. Test vague and highly specific questions.
4. Add Arabic documents and questions.
5. Replace TF-IDF with document embeddings.
6. Create a confidence rule from the highest similarity score.
7. Evaluate one successful and one failed query.

9. مهام الطالب

1. إضافة 20 مقطعًا هندسيًا.
2. مقارنة قيم topK.
3. اختبار أسئلة عامة ودقيقة.
4. إضافة مستندات وأسئلة عربية.
5. استبدال TF-IDF بتمثيلات مستندات.
6. إنشاء قاعدة ثقة اعتمادًا على التشابه.
7. تقييم حالة نجاح وحالة فشل.



Research Extension

Course assistant: index all lecture HTML pages, retrieve relevant sections, generate bilingual answers with citations, and compare sparse TF-IDF retrieval with dense embeddings and hybrid search.

امتداد بحثي

أنشئ مساعدًا للمقرر يفهرس جميع صفحات HTML ويجيب باللغتين مع الإحالات، ثم قارن الاسترجاع المتناثر والكثيف والهجين.



Review Questions

1. What is the role of TF-IDF?
2. How is cosine similarity computed?
3. Why is top-k retrieval needed?
4. What is grounding?
5. Why must retrieval and generation be evaluated separately?
6. What causes an unanswerable query?
7. How can dense embeddings improve retrieval?

أسئلة المراجعة

1. ما دور TF-IDF؟
2. كيف يحسب تشابه جيب التمام؟
3. لماذا نستخدم top-k؟
4. ما معنى الاستناد إلى المصادر؟
5. لماذا نقيم الاسترجاع والتوليد بشكل منفصل؟
6. ما سبب عدم قابلية بعض الأسئلة للإجابة؟
7. كيف تحسن التمثيلات الكثيفة الاسترجاع؟

المراجع | References

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